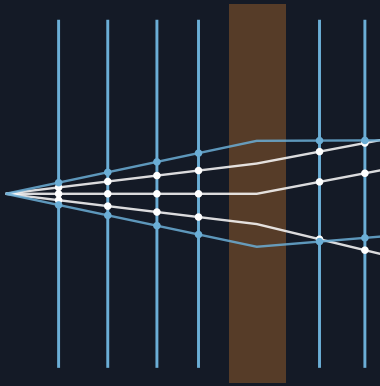
MUON BACKGROUND, DESIGN SPACE AND RECONSTRUCTION, LIVE

A veto for SHiP

Shah Rukh Qasim

University of Zurich · with G. Frisella

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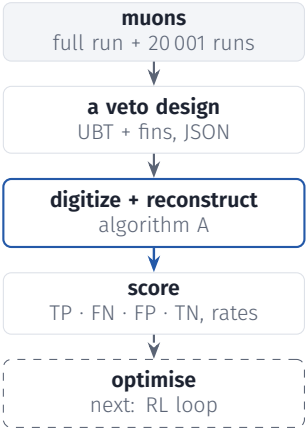
How to read this talk

Every plot with a slider is computed while you watch, by the code of the study (`veto/lib/veto`):

- the muon samples of the full simulation and the 20 001 old runs
- digitization, accidental hits and the veto reconstruction
- design comparisons, rerun on request

Slow widgets show a spinner and fill in; some show a quick answer first. 3D views rotate and zoom.

> The spill is **1 s** of $5 \cdot 10^{13}$ POT and the weights are muons per spill, so muons per spill and Hz are the same number.



PART 1

The problem

Muons from the target reach the detector; two of them can fake a decay in the vessel. The veto has to see them on their way in, without killing the signal.

SHiP and its muon background





Target at $z = 0$, **muon shield** (TRY6) to 30 m.

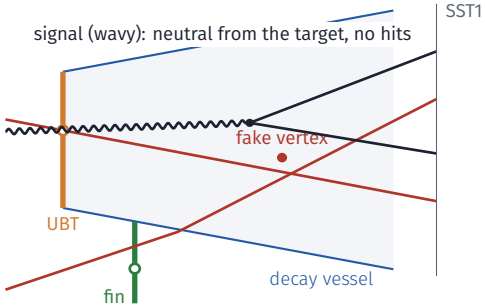
Decay vessel $z = 30.6$–80.6 m, 1×2.7 m entrance widening to 4×6 m.

UBT on the entrance window, **SST1** at 82 m, spectrometer magnet and timing detector behind.

Cavern wall at $x = -4.56$ m, floor at $y = -3.36$ m.

Muons per type (slider): bounced into SST1 · SST1 $p > 1$, timing · SST1 other · through an SBT wall · through the UBT window · bounced in the rock · straight past · stop in the rock · stop in the shield.

What the veto has to do



Two muons from the same spill, within ±200 ps, can form a vertex inside the vessel that points back to the target: a **fake signal**.

Each came in from outside: through the UBT window, the side walls, or after bouncing off the cavern. A hit where its track came from **explains** it.

muons per spill (simulated)	$1.8 \cdot 10^{10}$
crossing $z = 30.6$ m	$3.5 \cdot 10^{9}$
in the SST1 acceptance	$3 \cdot 10^{4}$
fake-vertex pairs, 15 years	**36** before any veto

> Veto every fake vertex, but do not veto the signal with **random hits** of the 10^{9}–10^{10} other muons.

PART 2

The data

Two simulations: 20 001 runs that keep only what reaches the spectrometer, and one run that keeps every muon.

Two simulations of the same muons

	20 001 old runs	**the full run** (`Full_dataset_z_3060_8200.pkl`)
muons	only those ending in the SST1 acceptance, ≈ 1 300 per run, 26.5 M rows	all 506 M simulated, one row each, 32 GB
states	start, z = 30.6 m, z = 82 m	z = 30.6 m and z = 82 m, or where the muon stopped
good for	pair background: ≈ $2 \cdot 10^5$ fake-vertex candidates	everything that misses the spectrometer: UBT and fin rates, veto-only muons
read with	a column store, 78 s to build	memory-mapped in place, a pass in ≈ 1 min

- The full run is a **rerun of seed 42**: same input muons, same states at 30.6 m; old file 42 holds its SST1 muons. Combined, it is not a 20 002nd run.
- `evt_n` = input-muon index in both; weights belong to the input muon (identical in both).
- Muons that bounce off the cavern differ between the two: bouncing is random from run to run.

Where the muons go



70 % stop in the shield; 26 M range out in the rock behind the cavern wall.

125 M pass the vessel to z = 82 m outside the spectrometer: harmless themselves, but they light up any detector placed in their way.

Only ≈ 2 000 per run matter directly: they enter the vessel or reach SST1.

> The veto sees 10^9–10^{10} muons per spill to catch ≈ 10^4. Accidental hits are the real enemy.

Where the muons go, in Hz



The same muons as rates: muons per spill of $5 \cdot 10^{13}$ POT over the 1 s spill.

≈3.5 GHz cross z = 30.6 m and 2.7 GHz pass the vessel to z = 82 m.

≈29 kHz reach the SST1 acceptance (the three SST1 bars together), ≈27 kHz enter the vessel without reaching it.

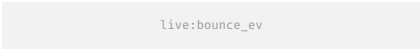
> A veto detector sees GHz in total; the muons it must catch arrive at tens of kHz.

THE DATA

Bouncing off the cavern







Grey dashed: straight line from the 30.6 m state; orange dotted: backward line from the 82 m state. They meet within 4 cm (median) at the bounce point, on the $-x$ wall or the floor. Right: all reconstructed bounce points trace the cavern.

The design-optimisation dataset

type	source	rows	muons / spill
SST1, bounced	old runs	6.0 M	$1.0 \cdot 10^4$
SST1, $p > 1$ GeV, timing	old runs	6.0 M	$8.2 \cdot 10^3$
SST1, other	old runs	14.5 M	$1.2 \cdot 10^4$
veto only	full run	975	$3.3 \cdot 10^4$
bounced in the rock	full run	575 k	$1.8 \cdot 10^7$
straight past	full run	15 M of 125 M	$2.7 \cdot 10^9$
stops in the rock	full run	15 M of 26 M	$7.7 \cdot 10^8$

`design-optimization-dataset-alpha-14-oct-2026`: 57 M muons, 5.8 GB.

Each type capped at 15 M rows and re-weighted, so `weight` is always muons per spill:

$$w = w_{\text{orig}} \cdot \frac{1}{N_{\text{runs}}} \cdot \frac{N_{\text{type}}}{N_{\text{kept}}}$$

Columns: states at 30.6 and 82 m, start state (old runs), bounce point, SBT walls crossed, type, run, `evt_n`.

Veto-only and rock-bounced come from one run: right rate, few distinct muons. A new run will fix that.

PART 3

The design space

A UBT on the vessel entrance and thin fins reaching out from the side walls, left and right. Nothing on top or below.

UBT and fins















Sliders: UBT top · fins per side · gap · fin length · fin top [m]. Fin colour: $\log_{10}$ muons/m^2/spill, 2 (purple) to 9 (yellow). Black: drawn muons crossing nothing. Bars: geometric coverage.

Where the muons are

live:rate_map

live:map_z

live:map_side

Muon density on a fin plane at z, every muon of the dataset; u is the distance from the vessel wall. Side: 0 right ($-x$), 1 left ($+x$). Dotted: cavern wall and floor limit.

A **lobe** at $y \approx 0$, 1.5–3 m from the wall, reaches 10^9 muons/m^2 per spill: the muons the shield sweeps past the vessel. Close to the wall and away from $y = 0$ it is 3–5 orders of magnitude quieter.

> A fin pays for every muon crossing it. The muons it is there for cross it near the wall.

Seven designs to learn from

		area
D0	UBT on the entrance only	3 m^2
D1	taller UBT, one 4 m fin per side at 30.6 m	56 m^2
D2	4 fins per side every 15 m, 3 m long, $y \pm$ 3 m	145 m^2
D3	7 fins on the right up to the wall, 3 short on the left	209 m^2
D4	8 fins per side, 5 m, y to 6 m: the stress test	624 m^2
D5	D2 with 50 cm or 5 cm cells	145 m^2

A design is a small JSON file: UBT edges and cell size; per fin its side, z, length, y range, cell size; detector efficiency and time resolution.

The loader applies the physical limits (floor, cavern wall, minimum UBT) and says what it changed.

All: 10 cm cells, $\varepsilon = 0.99$, $\sigma_t = 0.5$ ns, no tilt yet.

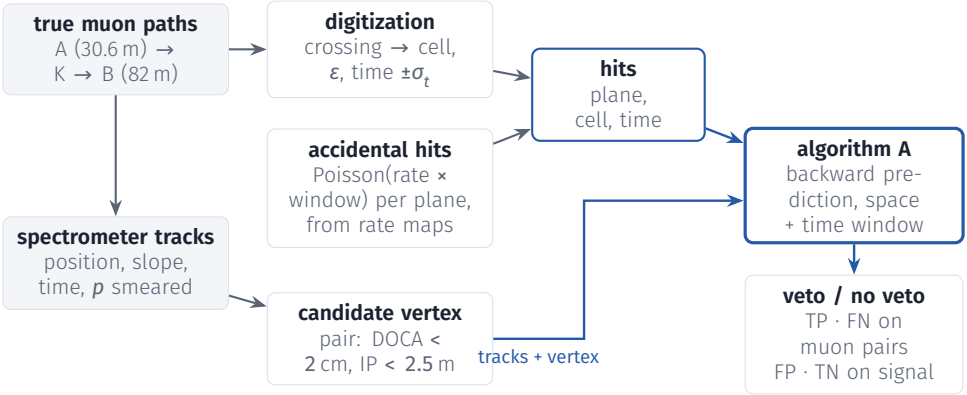
> In the RL environment fins are placed one after another along z with a minimum gap.

PART 4

Reconstruction

Events are short time windows around a spectrometer candidate. Each spectrometer track is followed back to the veto planes; a hit where it should be explains it.

The chain



Signal events: a long-lived neutral from the target decaying in the vessel into two tracks, with the pairs' selection; no own veto hits, only accidental hits can veto them.

Events are time windows, not spills





cell size [cm]; 1 s spill

A fin at 30.6 m sees ~ 10^9 muons per second: simulating whole spills hit by hit is hopeless.

Instead, around each candidate: the own muons' hits, plus per plane

$$n_{\text{acc}} \sim \text{Poisson}\left(R_{\text{plane}} \cdot \frac{2W}{T_{\text{spill}}}\right)$$

random hits, placed according to the full-statistics rate map and digitized like real ones.

Blue box: a search window near the wall. What matters is the chance a random hit lands in it: rate density × box × window.

Orange squares: random hits of one ±1.5 ns window on the right fin of D2, at the chosen cell size.

The detector and spectrometer models

Veto planes (UBT and every fin, the same model)

- one layer of cells at fixed z (10 cm default)
- a crossing makes one hit with probability ε (0.99)
- hit = cell centre + time, σ_t = 0.5 ns, muons at c
- no direction, no energy, no noise, no dead time (yet)
- a muon's own crossing comes from its true path, bounce included

Spectrometer (track at z = 82 m)

- position ±1 mm, time ±0.1 ns
- slope ±0.1 mrad ⊕ 2 mrad·(1 GeV/p)
- momentum as true, used only to size the window

Direction comes only from the spectrometer. The UBT gives no usable direction; neither do the fins.

Candidates

- background: opposite-charge muon pairs of the same old run, p > 1 GeV, in the timing acceptance, DOCA < 2 cm, vertex in the vessel, IP < 2.5 m: **196 736** pairs
- each weighted by $w_1 w_2 N_{\text{runs}} \cdot 2\Delta t / T_{\text{spill}}$, Δt = 200 ps
- signal: long-lived neutrals from the target, nearly along the beam ($\theta \approx p_T/p$), decaying in the vessel into two tracks; same selection as the pairs
- single SST1 muons (200 k) for per-muon efficiencies

The signal: from the target, not from Mars

live:sig_cross

A long-lived neutral from the **target**: p = 10–100 GeV (log-uniform), p_T ~ exp(1 GeV), so $\theta \approx p_T/p$.

Decays uniformly in z in the vessel, m = 1 GeV → two muon-mass tracks.

The pairs' selection: SST1, p > 1 GeV, timing, IP < 2.5 m. **37 %** pass; median daughter angle 22 mrad.

Orange: its tracks followed back to z = 30.6 m; grey: muon density.

> Two thirds cross the quiet UBT; the rest hit the fins next to the wall, where random hits veto them.

Algorithm A: follow every track back

For each spectrometer track:

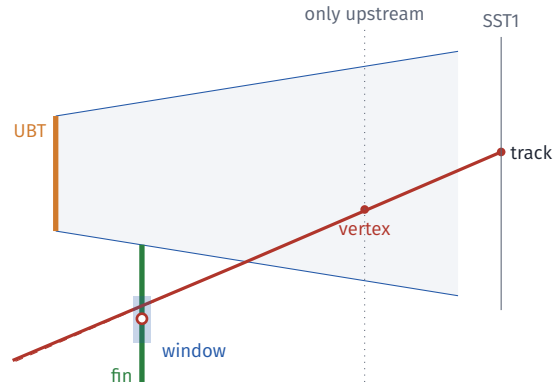
1. extrapolate backward in a straight line from $z = 82$ m (no field in the vessel)
2. on every veto plane **upstream of the vertex**: predicted cell (u, y) and time $t - \ell/c$
3. look for a hit within

$$|\Delta u|, |\Delta y| \le r_0 + \left(a + \frac{b}{p}\right) d + \tfrac{\text{cell}}{2}, \quad |\Delta t| \le n\,\sigma$$

4. a hit found: the track is **explained**

A candidate is vetoed if either track is explained.

> No combinatorics: one binary search per (event, plane) and a scan of that plane's few hits.



The spatial window

live:window_plot

live:r_ms

$r = r_0 + (a + b/p)\,d$ with $r_0 = 5$ cm, $a = 0.5$ mrad and b the slider, in mrad·GeV: the spectrometer's slope error grows with the lever arm d, multiple scattering with $1/p$.

Title: how many of the true crossings of straight SST1 muons (D2, 60 k muons) fall inside the window, computed now.

> For straight muons the prediction is within millimetres: the window can be tight. A bigger window only buys more random hits.

The time window

live:time_plot

live:time_trade

live:sig_t

live:n_sig

The predicted time includes the time of flight from the plane to z = 82 m. The window is $\pm n\sigma$ with $\sigma = \sigma_t^{\text{veto}} \oplus \sigma_t^{\text{track}}$: wider loses fewer own hits, but lets in proportionally more random hits.

Right: signal loss scaled from D2's 1.3 % at 3σ and 0.5 ns, since random hits grow with the window.

RECONSTRUCTION

Algorithm A on real events





design: 0 D0 · 1 D1 · 2 D2 · 3 D3 · 4 D4 · 5 D5 coarse · 6 D5 fine · 7 fibre · 8 pixel · 9 straw · 10 tile · 11 tile2 (fins). kind: 0 pair vetoed · 1 pair missed · 2 signal lost to a random hit · 3 bounced muon missed.

Solid: true path; dotted: backward prediction; white: own hits; diamond: the hit that explained the track; grey: random hits.

The same event from above



x–*z* view. Thick lines: UBT (orange) and fins (green). Blue bars: the search window on each plane the track is followed to. Same sliders as the previous slide.

What it costs

design	planes	hits / event	comparisons	µs / event
D0	1	0.1	40 k	0.5
D1	3	16	3.2 M	4.8
D2	9	39	4.0 M	12.5
D3	11	49	4.9 M	16.7
D4	17	90	7.2 M	33.3

196 736 background events; numpy, one core of a laptop.

A full evaluation of a design (196 k pairs, 100 k signal, 200 k single muons) takes **≈6 s** on one core, linear in hits per event.

Nothing to cap: algorithm A never combines hits.

> On a GPU, or with the factorised estimate, 10–100 ms per design: fast enough for an RL loop.

PART 5

Results

The background is easy to veto. Keeping the signal, and catching the last muon pairs, is where designs differ.

RESULTS

Seven designs



	D0	D1	D2	D3	D4	D5 coarse	D5 fine
background vetoed	25 %	99.992 %	99.997 %	99.992 %	100.000 %	99.997 %	99.997 %
pairs surviving	171 020	54	27	19	6	24	27
signal lost	0	1.13 %	1.28 %	1.49 %	1.76 %	4.72 %	1.00 %

RESULTS

Turn the knobs











design: 0 D0 · 1 D1 · 2 D2 · 3 D3 · 4 D4 · 5 D5 coarse · 6 D5 fine · 7 fibre · 8 pixel · 9 straw · 10 tile · 11 tile2 (fins). The whole chain reruns: a quick answer on 15 % of the events, then all of them (≈5--10 s).

Figure of merit

live:fom

live:fom_design

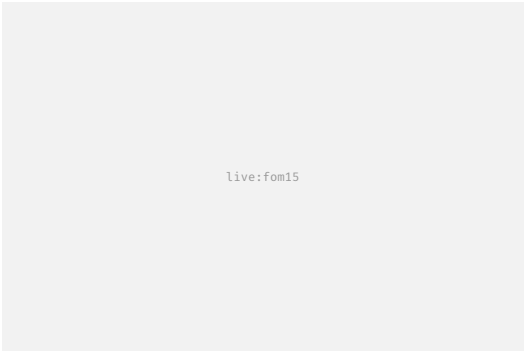
Positive = vetoed. Background: events in 15 years; signal: fractions.

design	TPR	FNR	FPR	TNR	B / 15 y	Punzi
D0	25.2 %	74.8 %	0	100 %	27	0.15
D1	99.992 %	0.008 %	1.13 %	98.87 %	0.003	0.637
D2	99.997 %	0.003 %	1.28 %	98.72 %	0.001	0.645
D3	99.992 %	0.008 %	1.49 %	98.51 %	0.003	0.634
D4	99.9998 %	0.0002 %	1.76 %	98.24 %	0.0001	0.651
D5 coarse	99.997 %	0.003 %	4.72 %	95.28 %	0.001	0.623
D5 fine	99.997 %	0.003 %	1.00 %	99.00 %	0.001	0.647

Punzi = $\varepsilon_S/(1.5 + \sqrt{B})$, ε_S = TNR. Every design with fins has $B \ll 1$, so the FoM is set by the signal efficiency.

> B rests on a few dozen surviving pairs and some muons weigh 100× more: one pair can move it by 5×. The factorised estimate comes next.

Figure of merit: the 15-year run







The same confusion matrix, as **counts over 15 years**: $6 \cdot 10^{20}$ POT in 1 s spills of $5 \cdot 10^{13}$ POT, $1.2 \cdot 10^{7}$ spills.

Fake vertices: 36 muon pairs over the whole run look like a decay in the vessel. The veto removes all but a tiny fraction: FN is the expected number of **fake discoveries**.

Signal: its total is model dependent; the slider sets how many selected decays SHiP would see. FP are lost to random hits, TN kept.

$P(\geq 1 \text{ fake}) = 1 - e^{-\text{FN}}$: the chance the whole run has even one.

> Background-free means FN $\ll$ 1: true for every design with fins. They differ in how much signal they keep.

RESULTS

Experiments

variation of D2	FPR	TPR	B / 15 y
as is	1.21 %	99.997 %	0.001
σ_t 0.5 → **0.1 ns**	**0.41 %**	99.98 %	0.006
σ_t 1 ns	2.19 %	99.997 %	0.001
cells 10 → 3 cm	0.83 %	99.997 %	0.001
σ_t 0.1 ns **and** 5 cm cells	**0.30 %**	99.98 %	0.006
time window 3 → 2σ	0.89 %	99.91 %	0.03
time window 4σ	1.65 %	99.998 %	0.0008
veto needs hits on 2 planes	0.002 %	36 %	23
without the fins at 30.6 m	0.13 %	61 %	14
fins only 1 m long	0.20 %	98.9 %	0.4
fins only y = −1..1 m	1.21 %	94.2 %	2.1
D4, σ_t 0.1 ns and 5 cm cells	0.49 %	99.9998 %	0.0001

60 k signal decays per row; `bin/run_reco.py` on one core.

Time resolution is the biggest lever on signal loss, then cell size: 3–4× less, at no cost in background.

The **fins at 30.6 m** veto most of the background and cause most of the signal loss.

A **2-plane rule** fails: most muons cross exactly one plane. Redundancy has to be built into the geometry.

> Short fins trade 0.2 % signal loss for about 0.4 background events: the kind of choice the optimiser has to make.

Why the last pairs survive

live:missed_why

live:mw_design

A pair survives only if **neither** muon is explained. For a straight muon crossing a single plane that is one chance:

$$P_{\text{miss}} \approx (1 - \varepsilon) + 2\,\Phi(-n) \approx 1.3\,\%$$

so a pair escapes with $\approx 2 \cdot 10^{-4}$: ≈ 30–40 of 196 k.

Bounced muons are followed back only after their bounce: the prediction misses the planes they crossed before it by metres.

> Redundancy: two planes on every incoming path square the miss probability.

Bounced muons, in perspective

background pairs	count	weight
no bounced muon	193 873	98.7 %
one bounced muon	2 863	1.3 %
two bounced muons	0	0 %

Single SST1 muons vetoed by algorithm A:

	D2	D3	D4
p > 1 GeV, timing	99.1 %	99.2 %	99.3 %
other	99.3 %	99.7 %	100.0 %
bounced	74 %	87 %	88 %

Bouncing costs most of the momentum (median 3 → 0.3 GeV), so bounced muons rarely pass p > 1 GeV: 23 % of SST1 muons, 1.5 % of the pairs.

For single muons algorithm A's blind spot is real; for the pair background it is a small effect, and the partner muon usually explains the pair.

> A hit-driven algorithm B could use the hits before the bounce: tracklets in the fins pointing at the vessel.

Granularity: the signal-loss knob

live:cell_scan

live:cs_design

Same geometry, cell size from 3 to 50 cm, 30 k signal events per point.

Smaller cells shrink the search box, so fewer random hits fall in it: signal loss drops from 4.7 % (50 cm) to 1.0 % (5 cm) for D2.

But the busiest 10 cm cell already sees **60 MHz**, the UBT 400 Hz. Dead time and pile-up are not modelled yet and will matter in the lobe.

> Detector technology near $|x| \approx 2$ m is a design question of its own.

PART 6

Next

What the reconstruction still ignores, and how the design loop will use it.

NEXT

What is not in yet

Reconstruction

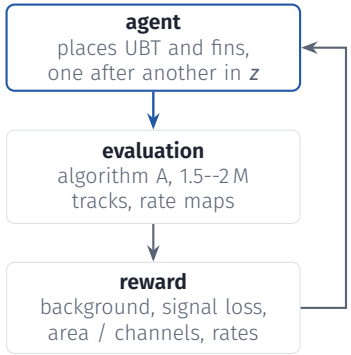
- algorithm B: hit-driven tracklets across fins (sees bounces), with a time cap per event
- factorised background: $\sum w P^{(1)}_{\text{miss}} P^{(2)}_{\text{miss}}$ instead of counting a handful of survivors
- dead time and pile-up at MHz rates per cell
- tilted fins (geometry and hits exist, not in the reconstruction)

Physics and data

- muon DIS in the walls and fins, neutrinos
- muons from the same proton collision (correlated in time)
- specific signal models (masses, decay modes, lifetimes) beyond the generic two-body decay
- more runs of the veto-only muons: per seed, everything crossing the vessel walls or UBT window, plus a 10^{-5} sample of UBT-plane crossers

NEXT

Towards the optimisation loop



Per design: ≈1 M SST1 muons, $2 \cdot 10^5$ pairs, 10^5 signal decays, and **no** flux muons as particles: they enter as rate maps.

Accidental vetoes analytic, $1 - e^{-\sum \text{rate} \times \text{window}}$: no Monte-Carlo noise in the reward.

> Expect the agent to cover incoming paths twice near the wall and to stay out of the lobe.

NEXT

Open questions

- POT per spill: are the weights muons per 1 s spill of $5 \cdot 10^{13}$ POT?
- The repeated-`evt_n` weight correction of the original analysis: the dataset uses the plain average over runs.
- Cavern geometry in the simulation: muons cross $x = -4.56$ m and reach $x \approx -9$ m.
- Veto technology near the vessel: rates per cell, time resolution, dead time.
- The new run: what to store per seed (above), and keep the start state and `evt_n`.

NEXT

Summary

Data: the full run and 20 001 old runs, combined into a 57 M-muon design dataset with per-spill weights; cavern bounces reconstructed to 4 cm.

Design space: a UBT plus fins from the side walls; seven designs from minimal to dense.

Reconstruction: time-window events with accidental hits from full-statistics rate maps; a track-driven veto, ≈10 µs per event, no combinatorics.

> Background is vetoed at > 99.99 % by any design with fins ($B \ll 1$ in 15 years). The figure of merit is then the signal efficiency: 98.7 % for D2, 99.7 % with 0.1 ns timing and 5 cm cells.